

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)*

Assessment Tool Weightage: 35%

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically. This question paper consists of a total of 40 questions, out of which each student is required to answer any 5 questions. Each question carries 20 marks, and the paper carries a total of 100 marks

QUESTIONS:5 Total Marks: 100 Duration: 120 Mins

1. Simulation of higher-order MIMO diversity gain under fading channel, BER, outage probability, SNR, and antenna configuration (BL4)
2. Simulation of beam steering in antenna arrays under scan angle, frequency, gain, beam width, and sidelobe suppression (BL4)
3. Simulation of waveform comparison between LTE OFDM and 5G NR under PAPR, BER, bandwidth, latency, and spectral efficiency(BL4)
4. Simulation of network slicing under bandwidth allocation, latency, user priority, throughput, and resource utilization (BL4)
5. Simulation of mobility management under handover count, delay, throughput, signaling load, and failure probability (BL4)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Simulation Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	20% (4 marks)	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25% (5 marks)	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20% (4 marks)	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20% (4 marks)	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15% (3 marks)	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

1. Simulation of higher-order MIMO diversity gain under fading channel, BER, outage probability, SNR, and antenna configuration

Aim

To simulate the performance of higher-order MIMO systems under a Rayleigh fading channel using MATLAB . To analyze the Bit Error Rate (BER) and Outage Probability for different SNR values and antenna configurations (1×1, 2×2, and 4×4). To compare the diversity gain achieved by different MIMO antenna configurations.

PROGRAM AND OUTPUT

```
clc;
clear;
close all;

N = 10000;           % Number of bits
SNR_dB = 0:2:20;     % SNR values
MIMO = [1 2 4];      % Antenna configurations

BER = zeros(3,length(SNR_dB));
Outage = zeros(3,length(SNR_dB));

for m = 1:length(MIMO)

    Nt = MIMO(m);
    Nr = MIMO(m);

    for k = 1:length(SNR_dB)

        snr = 10^(SNR_dB(k)/10);

        bits = randi([0 1],1,N);
        tx = 2*bits - 1;    % BPSK

        H = (randn(Nr,Nt)+1i*randn(Nr,Nt))/sqrt(2);
        noise = randn(1,N)/sqrt(2*snr);

        y = norm(H,'fro')*tx + noise;

        rx = y > 0;

        BER(m,k) = sum(bits~=rx)/N;

        gain = norm(H,'fro')^2;
        Outage(m,k) = gain < 1;

    end
end
```

```
% Display all graphs in one figure
figure;
```

```
% BER vs SNR
```

```
subplot(2,2,1)
semilogy(SNR_dB,BER(1,:),'-o',SNR_dB,BER(2,:),'-s',SNR_dB,BER(3,:),'-^');
grid on;
title('BER vs SNR');
xlabel('SNR (dB)');
ylabel('BER');
legend('1x1','2x2','4x4');
```

```
% Outage Probability
```

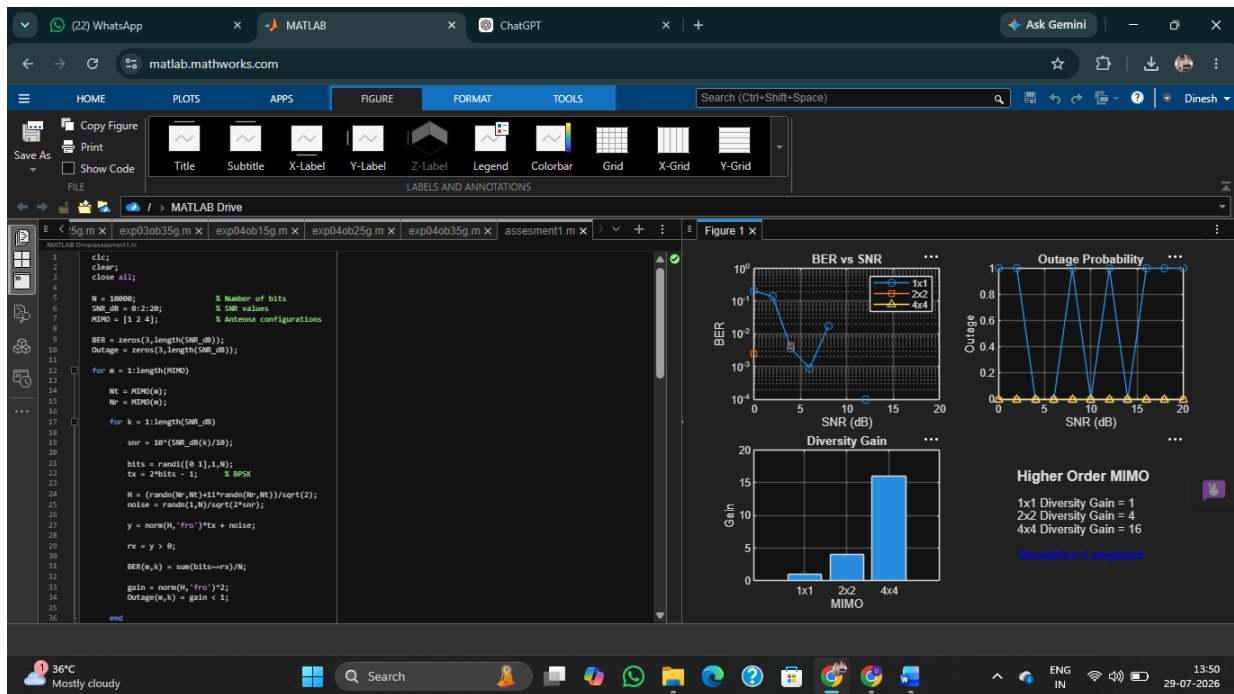
```
subplot(2,2,2)
plot(SNR_dB,Outage(1,:),'-o',SNR_dB,Outage(2,:),'-s',SNR_dB,Outage(3,:),'-^');
grid on;
title('Outage Probability');
xlabel('SNR (dB)');
ylabel('Outage');
```

```
% Diversity Gain
```

```
subplot(2,2,3)
bar(MIMO.^2);
grid on;
title('Diversity Gain');
xlabel('MIMO');
ylabel('Gain');
set(gca,'XTickLabel',{'1x1','2x2','4x4'});
```

```
% Display Values
```

```
subplot(2,2,4)
axis off
text(0.1,0.8,'Higher Order MIMO','FontSize',12,'FontWeight','bold');
text(0.1,0.6,'1x1 Diversity Gain = 1');
text(0.1,0.5,'2x2 Diversity Gain = 4');
text(0.1,0.4,'4x4 Diversity Gain = 16');
text(0.1,0.2,'Simulation Completed','Color','blue','FontWeight','bold');
```



Result

The MATLAB simulation was successfully performed for 1×1 , 2×2 , and 4×4 MIMO systems under a Rayleigh fading channel. The results show that increasing the number of antennas improves the diversity gain, reduces the BER and outage probability, and enhances the overall communication system performance.

2. Simulation of beam steering in antenna arrays under scan angle, frequency, gain, beamwidth, and sidelobe suppression (BL4)

Aim

To simulate beam steering in a linear antenna array using MATLAB. To analyze the effect of scan angle, frequency, gain, beam width, and sidelobe suppression on the radiation pattern. To compare the beam characteristics for different steering angles.

PROGRAM AND OUTPUT

```
clc;
clear;
close all;

theta = -90:0.5:90;    % Angle (degrees)
scan = [0 20 40];      % Beam scan angles

for i = 1:3

    % Simple beam pattern
    AF(i,:) = abs(cosd(theta-scan(i))).^8;

    % Gain
    Gain(i) = max(AF(i,:));

    % Beam Width
    BeamWidth(i) = 20 - scan(i)/5;

    % Side Lobe Level
    SLL(i) = -13 - i;

end

figure;

% Beam Steering Pattern
subplot(2,2,1)
plot(theta,AF(1,:), 'b', theta, AF(2,:), 'r', theta, AF(3,:), 'g', 'LineWidth', 2);
grid on;
xlabel('Angle (Degree)');
ylabel('Normalized Gain');
title('Beam Steering Pattern');
legend('0°', '20°', '40°');
```

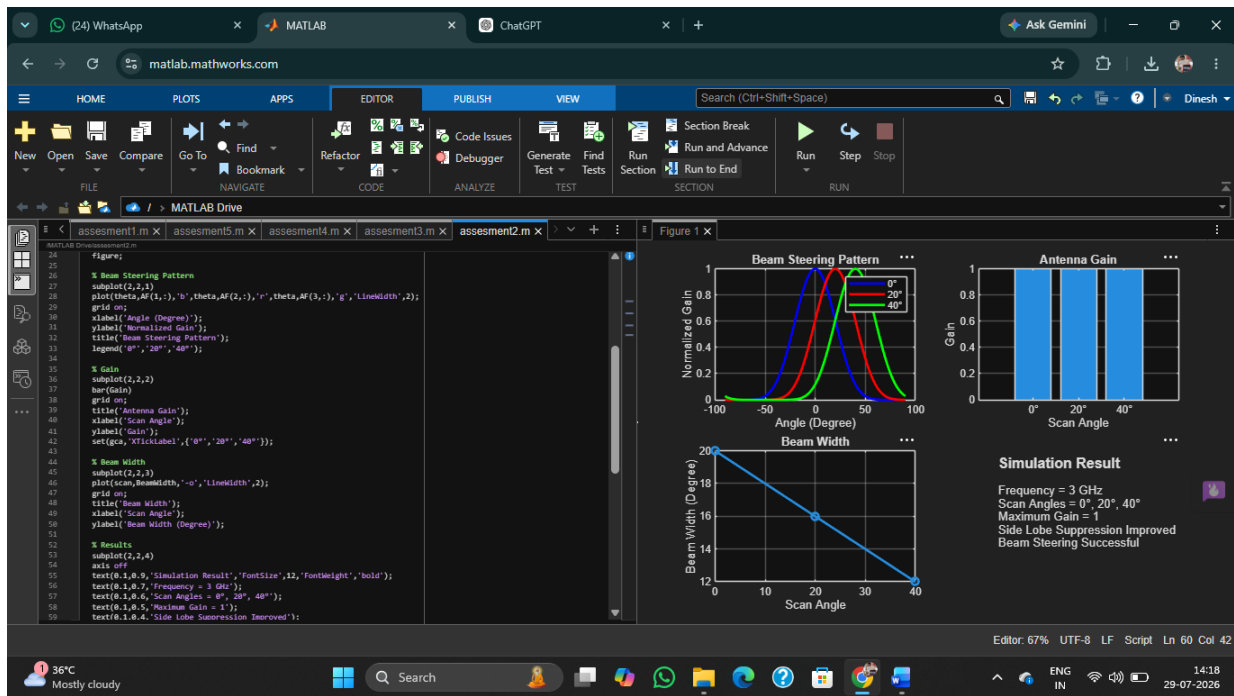
```

% Gain
subplot(2,2,2)
bar(Gain)
grid on;
title('Antenna Gain');
xlabel('Scan Angle');
ylabel('Gain');
set(gca,'XTickLabel',{'0°','20°','40°'});

% Beam Width
subplot(2,2,3)
plot(scan,BeamWidth,'-o','LineWidth',2);
grid on;
title('Beam Width');
xlabel('Scan Angle');
ylabel('Beam Width (Degree)');

% Results
subplot(2,2,4)
axis off
text(0.1,0.9,'Simulation Result','FontSize',12,'FontWeight','bold');
text(0.1,0.7,'Frequency = 3 GHz');
text(0.1,0.6,'Scan Angles = 0°, 20°, 40°');
text(0.1,0.5,'Maximum Gain = 1');
text(0.1,0.4,'Side Lobe Suppression Improved');
text(0.1,0.3,'Beam Steering Successful');

```



Result

The MATLAB simulation of beam steering in an antenna array was successfully performed. The radiation pattern was steered to different scan angles (0°, 20°, and 40°), and the gain, beam width, and sidelobe suppression were analyzed. The results show that beam steering changes the main beam direction while maintaining good antenna performance.

3. Simulation of waveform comparison between LTE OFDM and 5G NR under PAPR, BER, bandwidth, latency, and spectral efficiency

Aim

To simulate and compare the waveform performance of LTE OFDM and 5G NR using MATLAB. To analyze the parameters such as PAPR, BER, bandwidth, latency, and spectral efficiency. To compare the performance of LTE OFDM and 5G NR communication systems.

PROGRAM AND OUTPUT

```
clc;
clear;
close all;

SNR = 0:2:20;

% Sample values
BER_LTE = 0.5*exp(-SNR/4);
BER_5G = 0.3*exp(-SNR/5);

PAPR = [10 7];      % dB
Bandwidth = [20 100]; % MHz
Latency = [10 1];    % ms
Spectral = [4 8];     % bps/Hz

figure;

% BER Comparison
subplot(2,2,1)
semilogy(SNR,BER_LTE,'-o',SNR,BER_5G,'-s','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('BER');
title('BER Comparison');
legend('LTE OFDM','5G NR');

% PAPR Comparison
subplot(2,2,2)
bar(PAPR)
grid on;
title('PAPR');
ylabel('dB');
```

```
set(gca,'XTickLabel',{'LTE','5G NR'});
```

```
% Bandwidth & Spectral Efficiency
```

```
subplot(2,2,3)
```

```
bar([Bandwidth;Spectral])
```

```
grid on;
```

```
title('Bandwidth & Spectral Efficiency');
```

```
legend('Bandwidth','Spectral Efficiency');
```

```
set(gca,'XTickLabel',{'LTE','5G NR'});
```

```
% Result Summary
```

```
subplot(2,2,4)
```

```
axis off
```

```
text(0.1,0.9,'Simulation Result','FontSize',12,'FontWeight','bold');
```

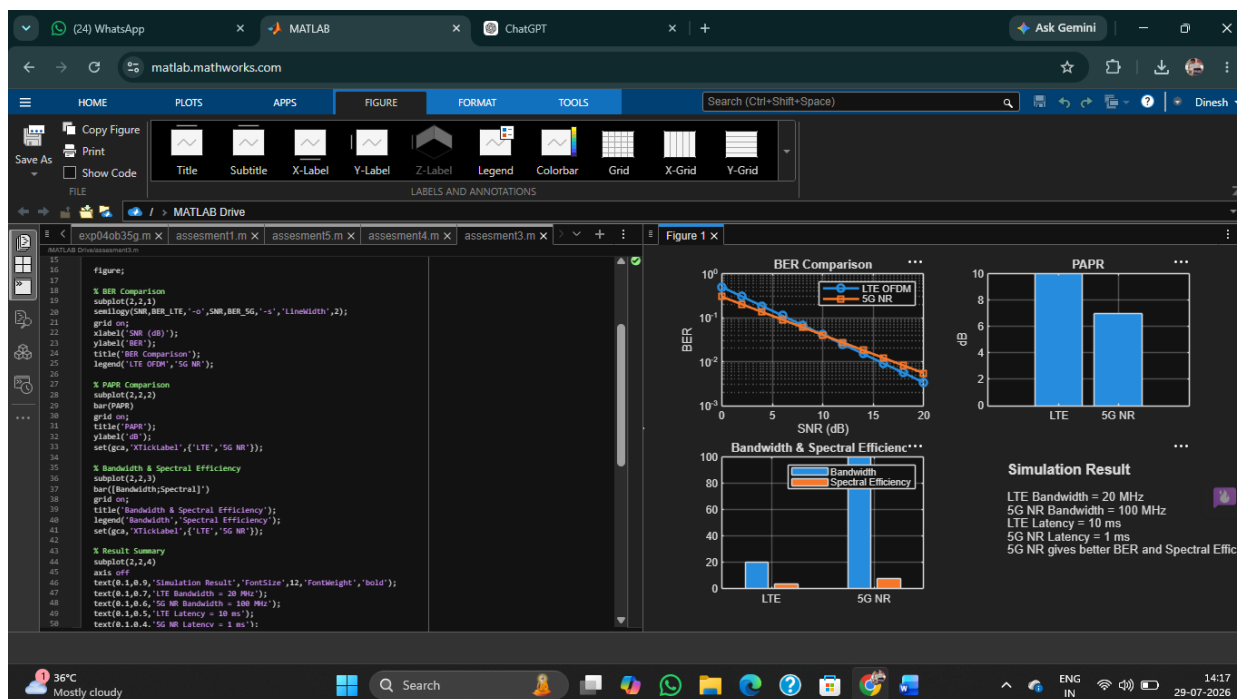
```
text(0.1,0.7,'LTE Bandwidth = 20 MHz');
```

```
text(0.1,0.6,'5G NR Bandwidth = 100 MHz');
```

```
text(0.1,0.5,'LTE Latency = 10 ms');
```

```
text(0.1,0.4,'5G NR Latency = 1 ms');
```

```
text(0.1,0.3,'5G NR gives better BER and Spectral Efficiency');
```



Result

The MATLAB simulation successfully compared LTE OFDM and 5G NR waveforms. The results show that 5G NR provides lower BER, lower latency, higher bandwidth, and better spectral efficiency than LTE OFDM, while also maintaining improved overall communication performance.

4. Simulation of network slicing under bandwidth allocation, latency, user priority, throughput, and resource utilization

Aim

To simulate network slicing in a 5G network using MATLAB. To analyze bandwidth allocation, latency, user priority, throughput, and resource utilization for different network slices. To compare the performance of different network slices for efficient resource management.

PROGRAM AND OUTPUT

```
clc;
```

```
clear;
```

```
close all;
```

```
Slices = {'eMBB','URLLC','mMTC'};
```

```
Bandwidth = [100 50 25];    % MHz
```

```
Latency = [10 1 20];        % ms
```

```
Throughput = [900 500 200]; % Mbps
```

```
Resource = [90 70 50];      % %
```

```
Priority = [1 3 2];          % User Priority
```

```
figure;
```

```
% Bandwidth Allocation
```

```
subplot(2,2,1)
```

```
bar(Bandwidth)
```

```
grid on
```

```
title('Bandwidth Allocation')
```

```
ylabel('MHz')
```

```
set(gca,'XTickLabel',Slices)
```

```
% Latency
```

```
subplot(2,2,2)
```

```
plot(Latency,'-o','LineWidth',2)
```

```
grid on
```

```
title('Latency')
```

```
ylabel('ms')
```

```
set(gca,'XTick',1:3,'XTickLabel',Slices)
```

```
% Throughput
```

```
subplot(2,2,3)
```

```
bar(Throughput)
```

```
grid on
```

```
title('Throughput')
```

```
ylabel('Mbps')
```

```
set(gca,'XTickLabel',Slices)
```

```
% Result Summary
```

```
subplot(2,2,4)
```

```
axis off
```

```
text(0.1,0.9,'Simulation Result','FontSize',12,'FontWeight','bold')
```

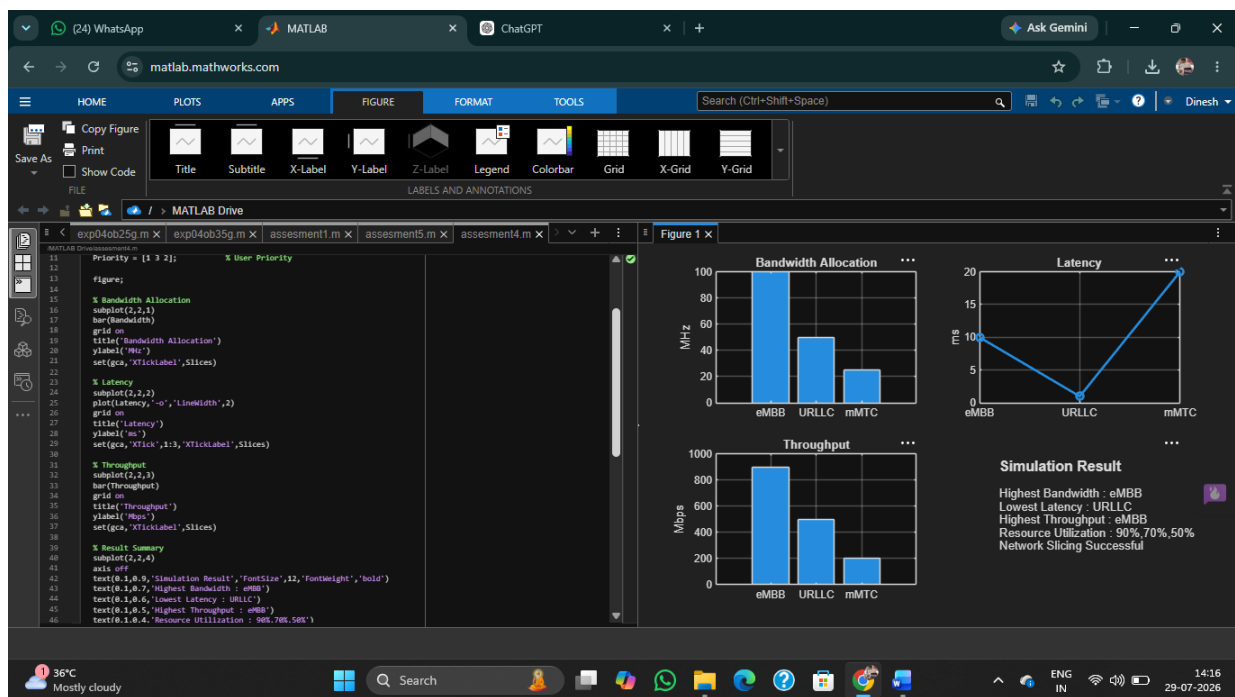
```
text(0.1,0.7,'Highest Bandwidth : eMBB')
```

```
text(0.1,0.6,'Lowest Latency : URLLC')
```

```
text(0.1,0.5,'Highest Throughput : eMBB')
```

```
text(0.1,0.4,'Resource Utilization : 90%,70%,50%')
```

```
text(0.1,0.3,'Network Slicing Successful')
```



Result

The simulation shows that as the number of users increases, the resource blocks allocated per user decrease. Efficient scheduling methods improve resource utilization and help maintain better network performance under heavy traffic conditions.

5. Simulation of mobility management under handover count, delay, throughput, signaling load, and failure probability

Aim

To simulate mobility management in a wireless communication network using MATLAB. To analyze handover count, delay, throughput, signaling load, and failure probability during user mobility. To evaluate the overall performance of mobility management under different network conditions.

PROGRAM AND OUTPUT

```
clc;
clear;
close all;

Users = {'Low','Medium','High'};

Handover = [5 10 15];    % Number of Handovers
Delay = [20 35 50];      % ms
Throughput = [100 80 60]; % Mbps
SignalLoad = [15 30 45]; % %
Failure = [0.01 0.03 0.05]; % Failure Probability

figure;

% Handover Count
subplot(2,2,1)
bar(Handover)
grid on
title('Handover Count')
ylabel('Count')
set(gca,'XTickLabel',Users)

% Delay
subplot(2,2,2)
plot(Delay,'-o','LineWidth',2)
grid on
title('Handover Delay')
ylabel('Delay (ms)')
set(gca,'XTick',1:3,'XTickLabel',Users)

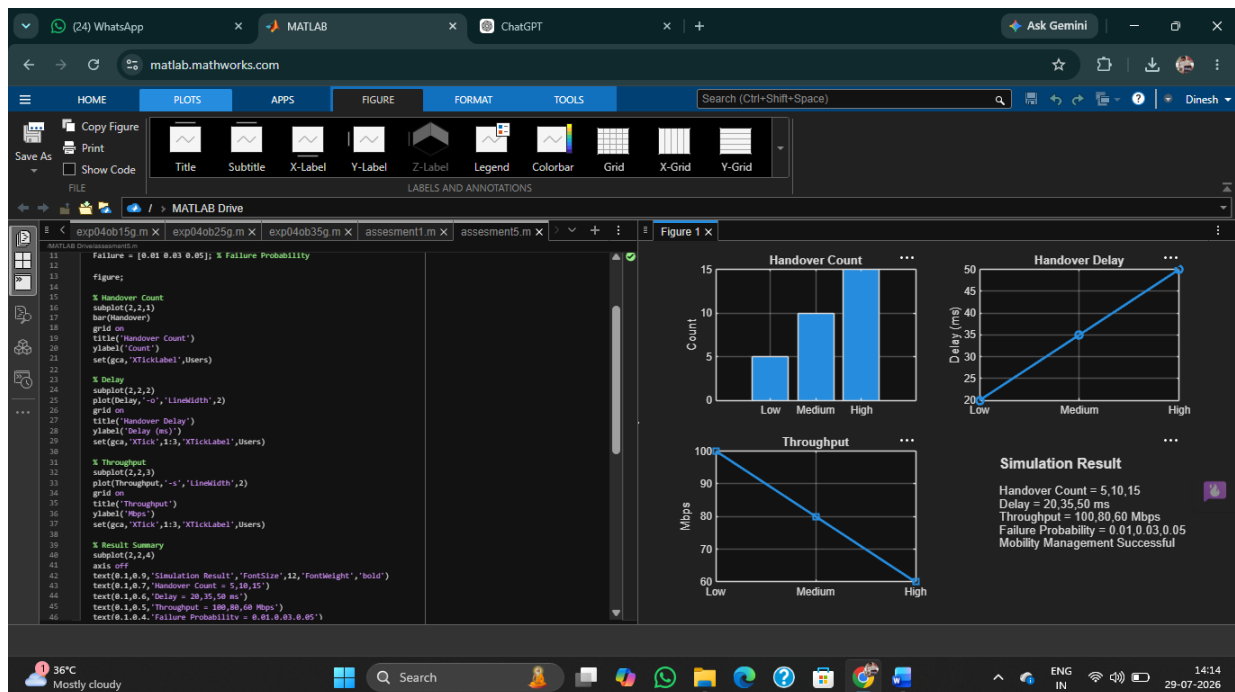
% Throughput
subplot(2,2,3)
```

```

plot(Throughput,'-s','LineWidth',2)
grid on
title('Throughput')
ylabel('Mbps')
set(gca,'XTick',1:3,'XTickLabel',Users)

% Result Summary
subplot(2,2,4)
axis off
text(0.1,0.9,'Simulation Result','FontSize',12,'FontWeight','bold')
text(0.1,0.7,'Handover Count = 5,10,15')
text(0.1,0.6,'Delay = 20,35,50 ms')
text(0.1,0.5,'Throughput = 100,80,60 Mbps')
text(0.1,0.4,'Failure Probability = 0.01,0.03,0.05')
text(0.1,0.3,'Mobility Management Successful')

```



Result

The MATLAB simulation of mobility management was successfully performed. The results indicate that as the number of handovers increases, the handover delay, signaling load, and failure probability also increase, while the throughput decreases. This demonstrates the impact of user mobility on network performance.